ENI - IRAQ

ZUBAIR OIL FIELD DEVELOPMENT PROJECT

OPERATION & MAINTENANCE MANUAL

VOLUME 1 – ZUBAIR FIELD AND HAMMAR OVERVIEW

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Operation & Maintenance Manual – Volume 1
Zubair Field and Hammar Overview

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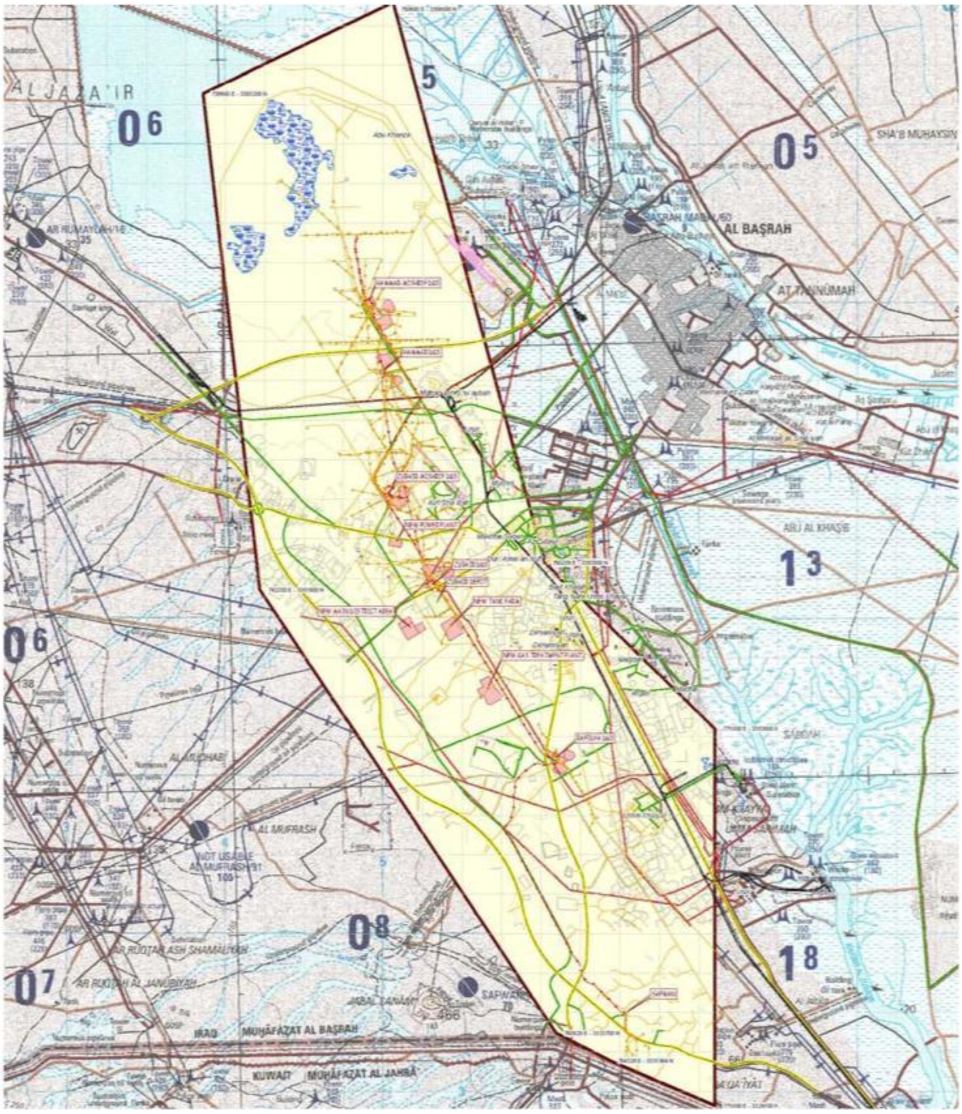
1.0 INTRODUCTION

1.1 Zubair Oil Field Development

The Zubair Oil Field Development project is for the development of the Zubair Oil Field in a joint venture with Occidental, Korea Gas and Missan Oil Company (as the state partner), and ENI as the operator. The Zubair Oil Field is located in the southern area of Iraq, 20km from Basra city (see Figure below). The structure of the Zubair Field is a relatively gentle anticline oriented NNW - SSE, approximately 60 km long and 10 to15 km wide.



New facilities are required in the Zubair Oil Field to increase the oil production capacity by an additional 300k BOPD. This will be achieved by installing six (6) Initial Production Facilities (IPFs), each sized to process and deliver 50k BOPD of stabilized crude to the export pipeline, plus associated gas and treated produced water.

Four (4) of the crude processing trains, with common associated gas compression, produced water treatment and utility systems are located at the new Hammar Initial Production Facility (IPF) site. One (1) crude processing train with associated gas compression, produced water treatment and utility systems is located at each of the new Zubair IPF site and Rafidiya IPF site. These new IPFs are installed near to the existing DGS at Hammar, Zubair and Rafidiya respectively.

1.2 Abbreviations

Project abbreviations used in the Hammar Operations and Maintenance Manuals are provided in Section 8.1 of this document.

1.3 Hammar Operations and Maintenance Manual

The full suite of Operation and Maintenance Manuals developed for the Hammar IPF are listed in Section 8.2 of this document.

Documents used during the development of this document are also listed in Section 8.2.

1.4 Hammar IPF Overview

The Hammar IPF site contains four crude processing trains with associated gas compression, produced water treatment, storage, utilities and safety systems.

Produced fluids are supplied to the Hammar IPF by four 18" flowlines, which are routed to the four processing trains via an inlet manifold.

The inlet manifold within the Hammar IPF battery limits allows limited crossover of the four feed lines from the Hammar DGS to the four crude processing trains within the Hammar IPF.

New manifolds feed the crude processing train within the IPFs from the nearby existing Degassing Station (DGS). These manifolds are installed within the existing DGSs and outside the Hammar IPF battery limits.

River water from the existing river water supply system is connected to the Hammar IPF by a 10" pipeline, which supplies water for the Potable Water, Wash Water, Service Water and Firewater Systems.

Crude oil from the Hammar IPF is routed to the export connecting pipeline via a 16" pipeline.

Raw gas from the Hammar IPF is routed to the main raw gas export line via an 18" pipeline.

Treatment water from the Hammar IPF for well injection is routed via an 18" pipeline fitted with a blind flange, which terminates at the Hammar IPF fence.

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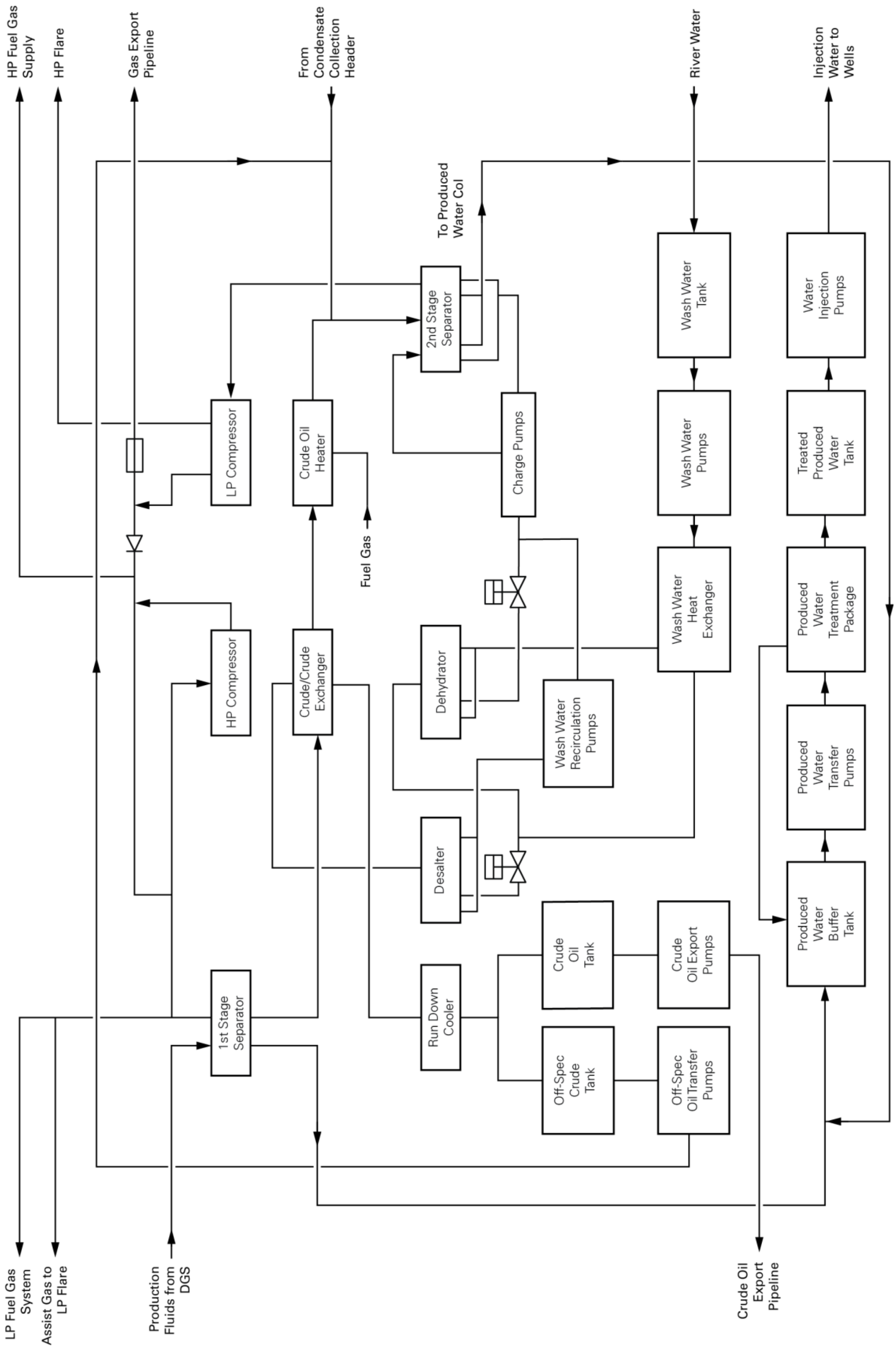

Figure 1 - Simplified Overview Hammar IPF

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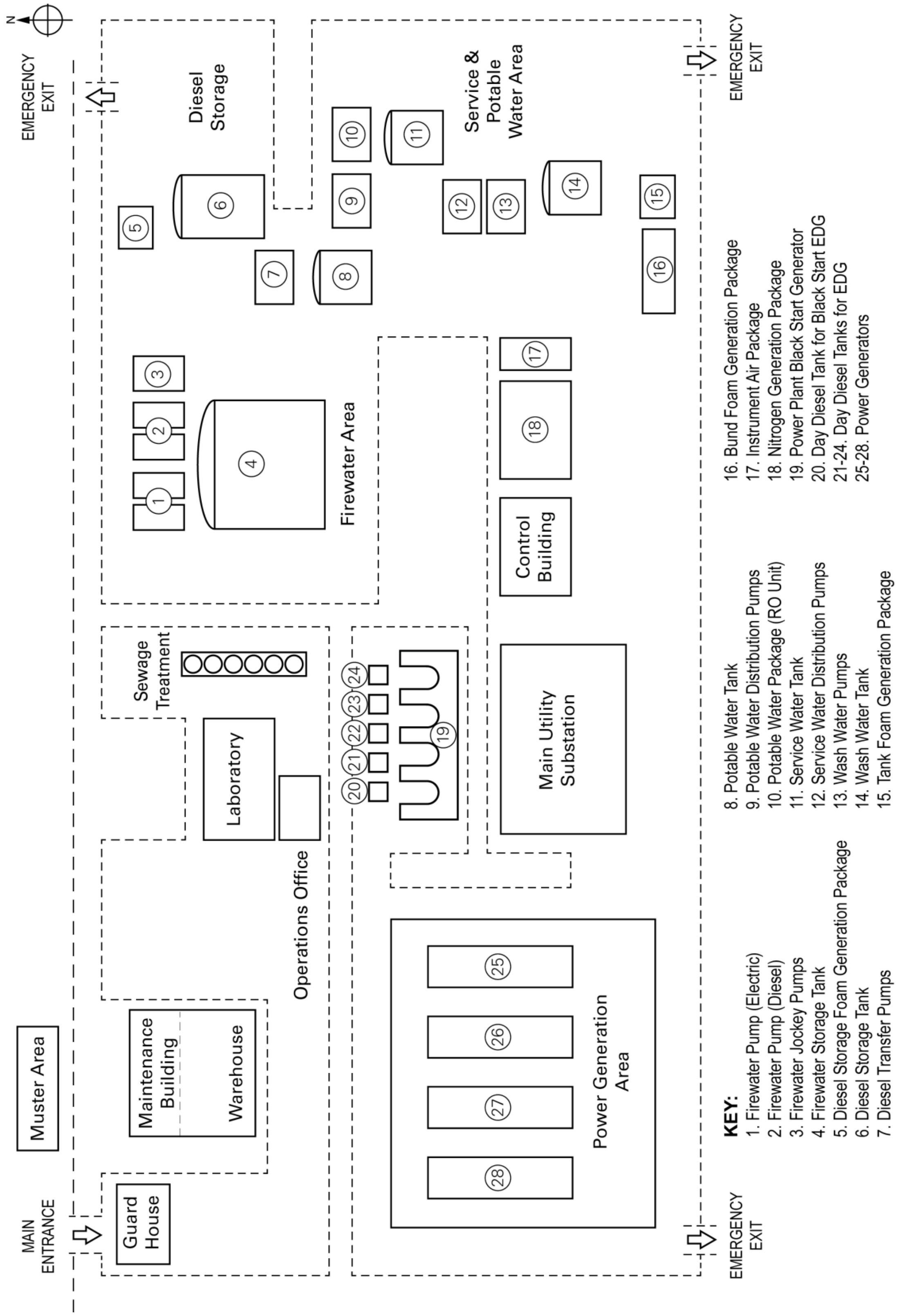

Figure 2 - Hammar IPF North West Area

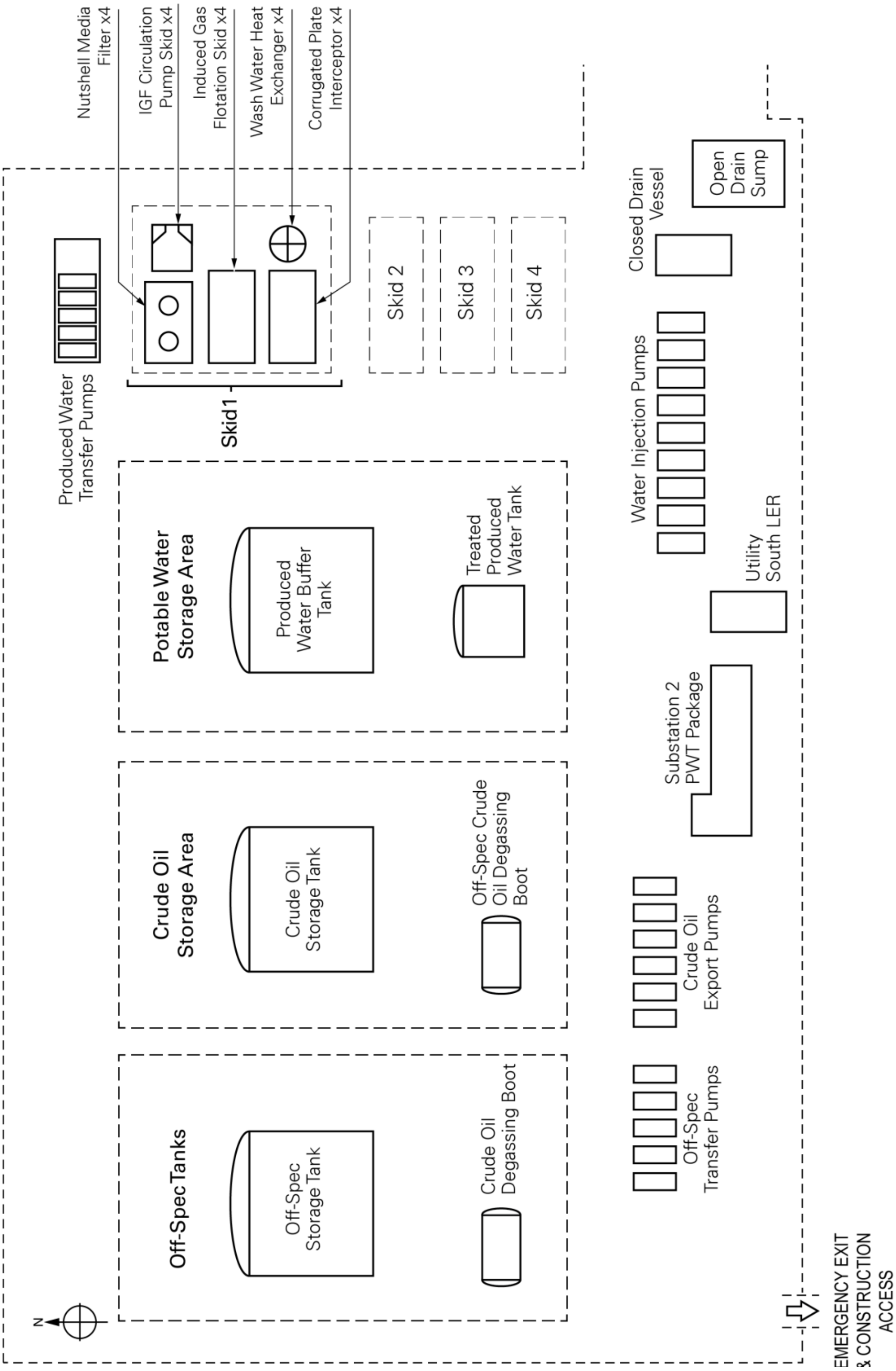

Figure 3 - Hammar IPF South West Area

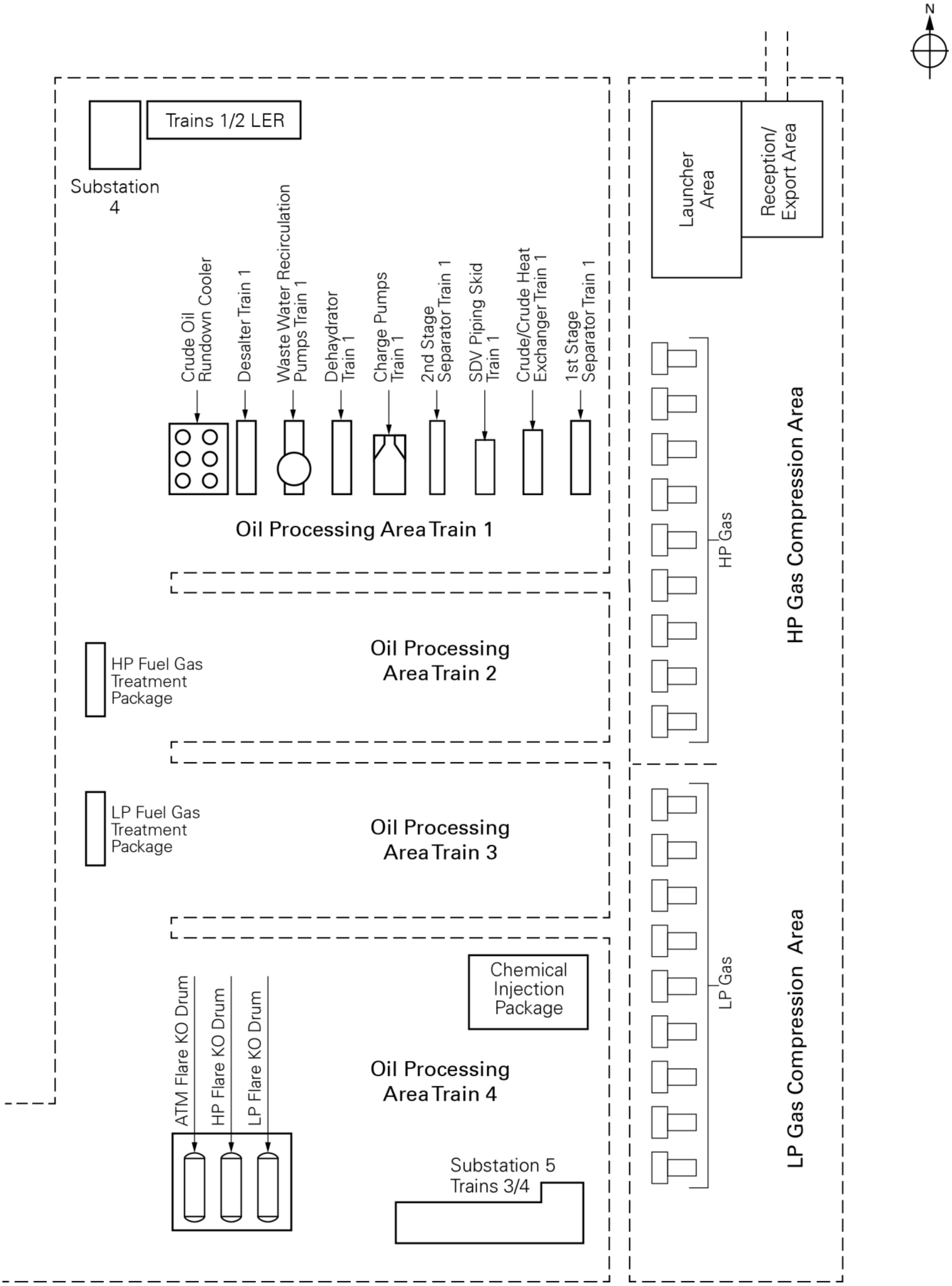

Figure 4 - Hammar IPF East Area

2.0 DESIGN SPECIFICATIONS

The systems and equipment at each IPF have been designed for and expected design life of 25 years including suitable material selection for the same.

2.1 Crude Fluid Feed Properties

For the process design basis, a single composition has been calculated for all three IPFs from a blend of Mishrif crude, 3rd Pay crude and 4th Pay crude, as shown below.

Component	Formula	Mol. Weight	Mole Fraction
Water	H_2O	18.02	-
Hydrogen Sulphide	H_2S	34.08	-
Nitrogen	N_2	28.01	0.004784
Carbon Dioxide	CO_2	44.01	0.008741
Methane	CH_4	16.04	0.398338
Ethane	C_2H_6	30.07	0.097434
Propane	C_3H_8	44.10	0.067954
i-Butane	$i\text{-}C_4H_{10}$	58.12	0.048801
n-Butane	$n\text{-}C_4H_{10}$	58.12	0.016458
i-Pentane	$i\text{-}C_5H_{12}$	72.15	0.023721
n-Pentane	$n\text{-}C_5H_{12}$	72.15	0.018444
Hexane	C_6H_{14}	86.18	0.030560
C7*	-	96.00	0.018893
C8*	-	107.00	0.017558
C9*	-	121.00	0.016352
C10-15*	-	165.68	0.077591
C16-20*	-	247.63	0.044327
C21-27*	-	327.85	0.041557
C28-35*	-	431.74	0.028833
C36-47*	-	565.42	0.022661
C48-80*	-	813.10	0.016993
Molecular Weight	-	-	104.51

Note:

1. Components marked with * are pseudo components.
2. The molecular weight of the crude fluid is a weighted average based on mole fractions.
3. A GOR value of 893 SCF/STOB has been used for gas facilities sizing.
4. The crude fluid feed is sweet, i.e., no hydrogen sulphide present.

2.2 Operating Conditions at the 1st Stage Separator Inlet

The expected operating conditions at the inlet of 1st stage separator are as follows:

Inlet Design Parameter	Units	Minimum	Maximum
Inlet feed fluid pressure	barg (psig)	15.2 (220)	36.6 (530)
Design pressure (pipeline and inlet manifold)	barg (psig)		138.7 (2012)
Inlet feed fluid temperature	°C	50 (winter)	77 (summer)
Design temperature (pipeline and inlet manifold)	°C	-5	107

2.3 Individual Formation Reservoir Dead Crude Oil Viscosities

The viscosity properties of the crude oils from various reservoirs are shown below:

Formation	Units	Mishrif	3rd Pay	4th Pay
Contribution	% Volume	46.0%	45.3%	8.7%
Viscosity 1 at temperature	cSt	33.47	16.21	18.85
	°C	15.6	15.6	15.6
Viscosity 2 at temperature	cSt	15.09	5.44	2.70
	°C	37.3	101.7	98.9

2.4 Individual Formation Reservoir Crude Fluid Compositions

The compositions of the Mishrif crude, 3rd Pay crude and 4th Pay crude as shown below.

Formation	Mishrif (Carbonate)	3rd Pay (Upper Sandstone)	4th Pay (Lower Sandstone)
Well Number	**ZB-32**	**ZU-6**	**ZB-148**
Component	**Mole Fraction**		
Water	--	--	--
Hydrogen Sulphide	--	--	--
Nitrogen	0.00579	0.004752	0.00193
Carbon Dioxide	0.00123	0.012099	0.01519
Methane	0.30327	0.388085	0.57429
Ethane	0.10562	0.092628	0.07772
Propane	0.07570	0.067576	0.04289
i-Butane	0.01604	0.055320	0.01011
n-Butane	0.04557	0.018675	0.03370
i-Pentane	0.02173	0.025121	0.01236
n-Pentane	0.02566	0.018309	0.01573
Hexane	0.03983	0.029944	0.01748

Formation	Mishrif (Carbonate)	3rd Pay (Upper Sandstone)	4th Pay (Lower Sandstone)
Well Number	**ZB-32**	**ZU-6**	**ZB-148**
Component	**Mole Fraction**		
C7*	0.02046	0.019926	0.01742
C8*	0.01931	0.018552	0.01589
C9*	0.01823	0.017272	0.01450
C10-15*	0.08978	0.081306	0.06390
C16-20*	0.05438	0.045631	0.03203
C21-27*	0.05402	0.041822	0.02609
C28-35*	0.04012	0.028057	0.01507
C36-47*	0.03416	0.020948	0.00929
C48-80*	0.02910	0.013976	00441
Molecular Weight	134.61	102.46	67.79

Note:

1. Components marked with * are pseudo components.

2. The Mishrif reservoir has the heaviest oil fraction.

3. The 4th Pay reservoir has the crude with the highest GOR.

2.5 Formation Water Properties

The expected composition and properties of the formation water from the reservoirs is as below:

Component	Units	Value
Sodium	ppm	58,676.6
Calcium	ppm	15,720.0
Magnesium	ppm	3,433.5
Potassium	ppm	2,277.3
Barium	ppm	0.0
Strontium	ppm	324.3
Iron	ppm	128.7
Chloride	ppm	131,989.0
Sulphate	ppm	500.0
Carbonate	ppm	0.0
Bicarbonate	ppm	102.5
Nitrate	ppm	0.0
Hydrogen Sulphide	ppm	0.00
Total Dissolved Solids	ppm	213,151.8
Property	**Units**	**Value**
pH	-	5.05
Specific Gravity	-	1.1354
Resistivity	Ohm.m	0.06405

2.6 Associated Sand Properties

The expected properties of the associated sand with the reservoir fluids are as below:

Sieve Size (µm)	Size Distribution	Cumulative Retained
500	12%	12%
300	29%	41%
150	51%	92%
125	1%	93%
90	0%	94%
75	5%	99%
Pan	1%	100%

The average density is 2,650 kg/m^3 and mean particle size 230 µm.

2.7 River Water Properties

The properties of the river water are as follows:

Component	Units	Value
Chlorine	ppm	3 to 5
Oxygen	ppm	6.5 to 8.3
Sodium	ppm	596
Calcium	ppm	304
Magnesium	ppm	122
Chloride	ppm	1036
Sulphate	ppm	876
Carbonate	ppm	0.3
Bicarbonate	ppm	223
Nitrate	ppm	0.0
Carbon Dioxide	ppm	11.7
Total Dissolved Solids	ppm	3044
Total Suspended Solids (TSS)	ppm	3 to 4
Total Hardness	ppm	1261
Total Alkalinity	ppm	183
Property	**Units**	**Value**
Supply Temperature	°C	6 min (winter) 40 max (summer)
Sample Temperature	°C	23
pH	-	7.5
Turbidity	NTU	1.2

2.8 Stabilised Oil Product Specification

The product specification for the stabilised product is as follows:

Parameter	Units	Value
Reid Vapour Pressure (RVP)	bara (psia)	≤ 0.76 (≤ 11)
True Vapour Pressure (TVP) at maximum ambient temperature of 50°C	bara (psia)	≤ 0.89 (≤ 12.9)
Basic Sediment & Water (BS&W)	% Volume	≤ 0.5
Salt Content	ppm Weight	< 30

2.9 Treated Water Specification for Water Injection

The treated water specification for injection into the wells is as follows:

Parameter	Units	Value
Oil Content	ppm volume	≤ 20
Suspended Solids – maximum size	µm	≤ 5
Maximum Oxygen Content	ppb weight	≤ 10

2.10 Diesel Specification

The specification for diesel fuel is as follows:

Component	Units	Value
Specific Gravity @ 15.6°C	-	0.85
Flash Point	°C	54 (minimum)
Pour Point	°C	-9 (maximum)
Viscosity @ 40°C	cSt	5.6
Sulphur Content	weight %	1.0 (maximum)
Rams – Bottom C.R. (on 10% Res)	weight %	0.2 (maximum)
Distilled @ 350°C	volume %	85 (minimum)
Diesel Index	-	55 (minimum)
Cetane N°	-	53 (minimum)
Calorific Value (Gross) EST	kJ/kg {kcal/kg}	45,217 (10,800)
Ash	weight %	0.01 (maximum)

3.0 UTILITY SYSTEMS OVERVIEW

3.1 Instrument Air

Note: For a detailed description of the Hammar IPF Instrument Air System, refer to Operation and Maintenance Manual Volume 2 – Document 00251121DWPV50003.

A single instrument air package is provided for the Hammar IPFs to supply utility air (for cleaning, maintenance pneumatic tools, etc.) and clean, dry instrument air for the control and shutdown systems and start air for HP and LP compressors.

The instrument air package includes the following main equipment:

- Air compressor units (2 x 100%)
- Air dryer units (2 x 100%)
- Coarse coalescer pre-filter
- Fine coalescer filter
- Heatless regenerative dryers
- Dust post-filter
- Instrument air receivers (2 x 50%).

Two air compressor units are provided: one on duty and one on standby. The unit on duty is started and stopped based on the air demand. If the unit on duty fails to start on demand, the unit on standby starts automatically.

The duty/standby selection of the compressors can be done manually or automatically by the unit control module (splitting running hours equally or in accordance with a pre-determined sequence).

Utility air is taken from the discharge of the air compressor units, upstream of the air dryer units.

Two air dryer units are provided: one on duty and one on standby. During normal operation, one unit is in operation while the other is isolated by closing the two XOVs at the inlet of the dryers.

The changeover between the duty and standby unit is done manually or automatically by the package control unit. On failure of the unit on duty, the standby unit is automatically lined-up and the other unit isolated. Each unit is also provided with manual isolation valves.

The two (2) air compressor units and the two (2) air dryer units are interconnected so that it is possible to feed any of the air dryer units from any of the air compressor units.

The instrument air receivers are sized to provide the maximum instrument air demand of the plant for at least one hour. The capacity required is based on the case of no flow from the air dryer units and a pressure change in the instrument air header from its maximum to its minimum operating pressure (10 to 4.5 barg).

A bypass is provided around the instrument air receivers so that it is possible to operate the instrument air package with the receivers out of service. However, it is not recommended to operate the plant with the air receivers bypassed for long periods.

3.2 Nitrogen System

Note: For a detailed description of the Hammar IPF Nitrogen System, refer to Operation and Maintenance Manual Volume 3 – Document 00251121DWPV50002.

The nitrogen generation package produces nitrogen for purging of equipment during start-up and for maintenance operations and as a back-up for tank blanketing and flare purge gas.

The nitrogen generation package includes the following main equipment:

Air compressor units, each one comprising:

- Wet air receiver
- Coarse coalescer filters (2 x 100%)
- Fine coalescer filters (2 x 100%)
- Superfine coalescer filters (2 x 100%)
- Electric heaters (2 x 50%)
- Membrane separation unit
- Nitrogen receivers (2 x 50%).

The nitrogen generation package is mainly used as back-up of fuel gas for tank and vessel blanketing and purge of flare headers and therefore, mainly used when fuel gas is not available, i.e., after plant shutdown.

During normal operation, nitrogen is used only for blanketing of the wash water tank, purge and snuffing of the crude oil heaters (during start-up and emergency) and in the utility stations for various maintenance work.

Due to the difference in nitrogen demand during plant shutdown and normal operation, the nitrogen generation package is designed to operate under two modes:

- High capacity mode, used after plant shutdown
- Low capacity mode, used during plant normal operation.

The Hammar IPF nitrogen generation package is provided with 5 x 20% air compressor units.

The air compressor units are started and stopped by the unit control panel based on the nitrogen demand.

In high capacity mode, all compressors are started simultaneously when required based on the plant demand for nitrogen. In low capacity mode, only one air compressor unit is started when required.

The plant main power generation system (gas turbine generators) has not been sized considering simultaneous operation of all air compressor units in the nitrogen generation package, since this is only required when fuel gas is not available and the plant is not in operation (nitrogen generation package in high capacity mode).

During high capacity mode, power is provided by the emergency diesel generators.

During normal operation, power to the nitrogen generation package (in low capacity mode) is provided by the plant main power generation system.

Each nitrogen generation package is provided with 2 x 100% filtration trains. One of the trains is in service, while the other is on standby. Changeover between the filtration trains is a manual operation, through the isolation valves at the inlet and outlet of each train.

Two electric heaters are provided for each nitrogen generation package. For high capacity mode, each heater provides half of the required duty (2 x 50%).

During low capacity mode, the whole heat demand can be provided by a single heater. Therefore, one of the heaters can be taken out of service for maintenance, if required. Manual isolation valves are provided and the inlet and outlet of each heater for this purpose.

The membrane separation unit comprises a total of 60 membranes for the Hammar IPF. This includes approximately 10% (6) spare membranes.

Each membrane is provided with an inlet and outlet isolation valve so that they can be individually isolated. During normal operation, spare membranes are isolated with their inlet and outlet isolation valves closed.

Changeover of duty/spare membranes is done manually.

During high capacity mode, all duty membranes (54) are required to operate.

During low capacity mode, only four (4) membranes are required.

Automatic on/off valves are provided to each nitrogen generator package to isolate the membranes that are not required during low capacity mode. These valves are opened during high capacity mode so that all the membranes required are lined-up.

The nitrogen generation package is provided with 2 x 50% nitrogen receivers. Two receivers are provided instead of a single one due to fabrication and transportation constrains.

The receivers provide the total nitrogen requirements for snuffing of one crude oil heater.

Nitrogen is stored in the receiver at 12 barg and let down to 6 barg, before being distributed to the users.

A bypass is provided around the nitrogen air receivers so that it is possible to operate the nitrogen generation package with the receivers out of service. However, it is not recommended to operate the plant with the nitrogen receivers bypassed for long periods.

3.3 Chemical Injection System

Note: For a detailed description of the Hammar IPF Chemical Injection System, refer to Operation and Maintenance Manual Volume 4 – Document 00251121DWPV50001.

Chemical injection is provided for the following chemicals:

- De-emulsifier (1st stage separators, 2nd stage separators, dehydrators and desalters)
- Corrosion inhibitor- gas phase (1st stage separators gas outlet, 2nd stage separators gas outlet, LP fuel gas treatment package, compressors gas outlet)
- Corrosion inhibitor- liquid phase (1st stage separators inlet, crude oil heaters outlet, dehydrators)
- Antifoam (1st stage separators, 2nd stage separators)
- Scale inhibitor (desalters, crude/crude exchangers inlet)
- Reverse emulsion breaker (produced water tank)
- Polyelectrolyte (PWT nutshell filters)
- Ferric chloride (PWT nutshell filters)
- Oxygen scavenger (wash water tank, injection water)
- Biocide (shock dosed only)

Chemical dosing facilities include the following main equipment:

- Storage tanks
- Injection pumps
- Skid accessories including flexible hoses, calibration tube, pressure safety valves, check valves and isolation valves for each pump
- IRCD valves and injection quills.

3.4 Diesel System

Note: For a detailed description of the Hammar IPF Diesel System, refer to Operation and Maintenance Manual Volume 5 – Document 00251121DWPV50004.

The Diesel System supplies diesel fuel to the diesel firewater pumps and diesel power generation units and include the following main equipment:

- Diesel storage tank
- Diesel transfer pumps (2 x 100%)
- Diesel filter (100 µm)

The diesel is offloaded from trucks using either the pumps provided by the truck, or using the diesel transfer pumps. When using the diesel transfer pumps there shall be no filling of the diesel day tanks and the diesel is pumped to the tank via the diesel filter.

3.5 Potable Water System

Note: For a detailed description of the Hammar IPF Potable Water System, refer to Operation and Maintenance Manual Volume 6 – Document 00251121DWPV50005.

A single Potable Water System at the Hammar IPF provides:

- A continuous supply of drinking water and seal water for the PW transfer pumps
- An intermittent supply of water for the eye wash stations, emergency showers, water injection pumps seals, IGF circulation pumps seals, evaporative cooling for gas turbines.

The Potable Water System includes following equipment:

- Potable water package (utilising reverse osmosis)
- Potable water tank
- Potable water pump system.

The water supply is provided from the incoming river water pipeline.

3.6 Service Water System

Note: For a detailed description of the Hammar IPF Service Water System, refer to Operation and Maintenance Manual Volume 7 – Document 00251121DWPV50006.

A single Service Water System is provided at the Hammar IPF site to supply water for non-continuous requirements, such as washing down of equipment, etc.

The Service Water System includes a service water tank and pumps.

The water supply is provided from the incoming river water pipeline.

3.7 Flare System

Note: For a detailed description of the Hammar IPF Flare System, refer to Operation and Maintenance Manual Volume 8 – Document 00251121DWPV50007.

Three systems, HP LP and Atmospheric (ATM) Flare, are provided at the Hammar IPF:

- HP flare – to collect releases gas from the 1st stage separators, and also the emergency release from the crude/crude heat exchangers, crude heaters and HP gas compressors
- LP flare - to collect emergency releases and blowdowns from the other pressurised systems and LP gas compressors
- Atmospheric (ATM) flare - to collect near atmospheric pressure releases from degassing boots, storage tanks, closed drain drum, compressors fuel gas PSVs and PWT packages.

The HP Flare System includes the following main equipment:

- HP flare KO drum
- HP flare KO drum pumps
- HP flare stack with purge seal
- HP flare tip, with pilot and spark ignition system.

The LP Flare System includes the following main equipment:

- LP flare KO drum
- LP flare KO drum pumps
- LP flare stack with purge seal
- LP flare tip, with pilot and spark ignition system
- Fuel gas assist – for smokeless flaring.

The ATM Flare System includes the following main equipment:

- ATM flare KO drum
- ATM flare KO drum pumps
- ATM flare stack with purge seal
- ATM flare tip, with pilot and spark ignition system
- Air blower for smokeless flaring.

There is a common flare control panel for the three (3) flares.

The flare headers are normally purged with fuel gas for start-up and to prevent air ingress in flare stacks during a shutdown. Nitrogen is provided as a back-up purge.

In addition, LPG bottles are provided for initial start-up and for pilot fuel during a shutdown.

The liquids collected in the flare KO drums are routed to the off-spec crude tank via the flare KO drum pumps, which operate in lead/lag mode.

Flow measurement is provided on each of the flare headers. In addition to measuring the flow, and total gas being flared, this flow is used to control the amount of assist gas to ensure that flaring is smokeless.

A CCTV System is provided for the monitoring of the flare operation and safety.

3.8 Drains System

Note: For a detailed description of the Hammar IPF Drain Systems, refer to Operation and Maintenance Manual Volume 9 – Document 00251121DWPV50008.

The Hammar IPF is provided with two separate Drain Systems: Closed Drain System (Unit 550) and Oil Contaminated Open Drain System (Unit 540).

3.8.1 Closed Drain System

The Closed Drain System collects liquid effluent from the process equipment and includes a closed drain drum and three (3) closed drain pumps.

There are no continuous feeds to the closed drain drum, which will only be used during planned maintenance operations. The drum is normally purged with fuel gas and nitrogen is provided as a backup.

The liquids collected in the closed drain drum are routed to the off-spec crude tank via the closed drain pumps, which operate in lead/lag mode.

Any gas released in the vessel is routed to the ATM flare system.

3.8.2 Oil Contaminated Open Drain System

The Oil Contaminated Open Drain System collects waste water contaminated with hydrocarbon liquids in normal operation or during routine operations, e.g., maintenance, and is collected from the process skid pans and tank bunded areas. This includes rainfall water and firewater from process skid pans and storage tank bunded areas.

Skid pans are provided for all process skid equipment, e.g., separators, pumps, exchangers and tanks sample points. Water effluent from the process skid pans and tank dyke area can be diverted to the oily water open drain.

Oily water flows by gravity through underground piping to a concrete chamber with two inlets. One inlet is from tank bunded areas and the other from the process skid pans. The chamber has two outlets. The bottom outlet feeds the CPI unit and the top outlet bypasses the CPI unit and is gravity fed to a seepage pit.

The CPI unit is used to separate the oil and sand from the water. As the oily water passes through the CPI unit, solids are accumulated at the bottom of the unit and oil floats to the top, where it is collected in an oil basket. The API type oil-water interceptor plate pack aids coalescence and removes the floating oil to the oil basket from where it is removed for disposal using trucks.

The solids accumulated in the separator are removed by the operator as required. Facilities and tools are provided so that the solids removal from the CPI unit can be carried out safely.

The treated water is discharged to oil contaminated open drain sump downstream of the CPI unit. The open drain pumps are used to empty the oil contaminated open drain sump into an overflow arrangement sump, which gravity feeds treated water to seepage pit.

During normal operation there is a small amount of continuous flow through the separator, which is mainly the discharge from the RO unit. The separator is sized to cope with a higher flow resulting from heavy rain or use of the fire deluge system.

During rain or fire deluge, oil contaminated water is diverted to the oily water separator for treatment. The oily water flows through the open drains and concrete chamber to the CPI unit. The oil water is treated in the CPI unit and the treated water is discharged to the oil contaminated open drain sump. The level within the oil contaminated open drain sump rises very quickly in a fire incident.

A level instrument within the oil contaminated open drain sump trips an automated valve on the feed line to the CPI unit. This results in a water level rise in the concrete chamber. The water from the chamber is then discharged via the bypass outlet straight to seepage.

Open drain pumps are arranged so that it is possible to take a sample from the oil contaminated open drain sump. Also, it is possible to dose the water in the oil contaminated open drain sump, if required.

The Oil Contaminated Open Drain System includes the following main equipment:

- Concrete chamber upstream to CPI separator
- Oil contaminated open drain CPI unit in a concrete pit
- Automated valve on feed to CPI unit from the concrete chamber
- Oil contaminated open drain sump with an overflow arrangement
- Open drain pumps with facility to take sample and dose water in oil contaminated open drain sump.

3.9 Fuel Gas System

Note: For a detailed description of the Hammar IPF Fuel Gas System, refer to Operation and Maintenance Manual Volume 10 – Document 00251121DWPV50009.

The Hammar IPF is provided with two fuel gas systems; a LP fuel gas treatment package and a HP fuel gas treatment package.

Feed to the LP fuel gas system is raw gas from the 1st stage separators. The LP fuel gas system produces LP and MP fuel gas.

LP fuel gas is used for flare pilots, flare and flare header purge, tank and vessel blanketing and crude oil heaters.

MP fuel gas is used in the gas engines of LP and HP gas compressors and in the PWT packages.

The LP fuel system includes the following main items:

- Fuel gas cooler (1 x 100% forced draft air cooler)
- LP fuel gas KO drum (1 x 100%)
- LP fuel gas heater (2 x 100%).

The purpose of the fuel gas cooler is to condense water and heavy hydrocarbons from the feed to the package. It is also provided to ensure that the temperature of the MP fuel gas to the gas compressor engines does not exceed the maximum value recommended by the package vendor (60°C). The cooler has been designed to reduce the temperature of the gas to 60°C in summer and 30°C in winter. These temperatures are based on ambient temperatures of 50°C and 20°C for summer and winter, respectively.

Under favourable ambient conditions, it may be possible to cool down the gas below the design values, which improve the quality of the fuel gas since more water and heavy hydrocarbons is be condensed. However, it is important to ensure that the gas is not cooled down below its hydrate formation temperature (10°C at normal operating pressure) to avoid problems with plugging by hydrates.

The outlet of the cooler is provided with a low temperature alarm, set 5°C above the hydrate formation temperature, i.e., 15°C, to alert operators of gas overcooling.

The cooler is equipped with manual louvers to allow operators to adjust outlet temperature and avoid overcooling during periods of low ambient temperature. Operators have also the option to turn off one of the fans, if required to avoid overcooling. A differential pressure transmitter is provided in equipment potentially affected by hydrate plugging, i.e., fuel gas cooler and LP fuel gas KO drum. This can be used to monitor the pressure drop and detect the onset of hydrate plugging problems.

MP fuel gas is taken upstream the LP fuel gas heaters and let down from 13.7 barg to 10.5 barg. The MP fuel gas is slightly superheated, approximately 2°C above its HC dewpoint. Unlike LP fuel gas, MP fuel gas is not heated up in the LP fuel gas heater due to the constraint in the maximum temperature of the fuel gas to the gas compressor gas engines imposed by the package vendor (60°C). Nevertheless, it has been confirmed that the gas engines of the HP and LP gas compressor can operate satisfactorily with this quality of MP fuel gas.

Each LP fuel gas package is provided with 2 x 100% LP fuel gas heaters with one in service and the other on standby. Changeover between the heaters is carried out manually by manipulation of the isolation valves at the inlet and outlet of each heater.

In the heaters, the temperature of the LP fuel gas is increased by approximately 6°C. After the heaters, the pressure of the LP fuel gas is reduced from 13.4 barg to 6.0 barg. Because of the pressure let down, the LP fuel gas is superheated by approximately 10°C.

The feed to the HP fuel gas system is either from the 1st stage separators (during HP operating mode) or from the discharge of the HP compressors (during LP operating mode).

HP fuel gas is used exclusively by the power generation system (gas turbine generators).

The HP Fuel Gas System includes the following main items:

- HP fuel gas KO drum (1 x 100%)
- HP fuel gas filter (2 x 100%)
- HP fuel gas heater (2 x 100%).

Each HP fuel gas package is provided with 2 x 100% HP fuel gas filters. One of the filters is in service and the other on standby. Changeover between the filters is carried out manually by the manipulation of the isolation valves at the inlet and outlet of each filter.

Two HP fuel gas heaters (2 x 100%) are provided to each HP fuel gas system. During normal operation, both heaters are on line, sharing the duty between them. In the event that one of the heaters is tripped, the remaining one is able to take over and provide all the required heating. If the tripped heater is not isolated immediately, cold gas continues flowing through it, bypassing the operating heater. In this case, the temperature of the gas at the outlet of the operating heater will increase so that the temperature of the gas after the mixing point with the cold gas is at its correct value.

In the heaters, the temperature of the HP fuel gas is increased so that the gas is superheated by at least 28°C at the inlet of the power generation system. The heaters are controlled so that differential temperature between the inlet of the heaters and the inlet of the fuel gas to the gas turbine generators is at least 28°C.

3.10 Power Generation

Note: For a detailed description of the Hammar IPF Power Generation System, refer to Operation and Maintenance Manual Volume 11 – Document 00251121DWPV50010.

The Electrical Power Generation System comprises the following main equipment:

- Main power generation units (4 x 33% gas turbines generators)
- Emergency diesel generators (4 x 25%)
- Black start generator (diesel engine)
- AC UPS

The Main Power Generation System provides the electric power to the plant during normal operation.

A basic load shedding system is provided that trips the non-essential high power loads (water injection pumps) in order to avoid shutdown the whole plant.

This system requires, however, that a minimum of three (3) gas turbine generators are kept in operation whenever the water injection pumps are in operation.

Generators provide the electric power to the essential and emergency systems in case of tripping/failure of the Main Power Generation System and also provide the power required to start the plant after a plant shutdown.

A single black start generator is provided that supplies the power required for the auxiliaries of one gas turbine generator during start-up.

3.11 Firewater System

Note: For a detailed description of the Hammar IPF Firewater System, refer to Operation and Maintenance Manual Volume 18 – Document 00251121DWPV50017.

The Firewater System on the Hammar IPF comprises the following main equipment:

- Firewater tank
- Diesel driven firewater pumps (3 x 50%)
- Electric motor driven firewater pump (2 x 100%)
- Electric motor driven fire water jockey pumps (2 x 100%),
- Deluge valve skids
- Tank foam generator packages
- Bund foam generator packages

- Water/foam hydrant/monitor
- Water sprinkler control assembly
- Multiple jet control (MJC) assembly
- Mobile foam units
- Surface mounted hose reels
- Portable ground monitors
- Fire extinguishers placed at strategic points around the Hammar IPF.

Water is routed from the existing treated river water system to the firewater tank under level control.

The firewater pumps transfer the firewater via pressure control to the firewater ring main and deluge valve skid, tank foam generator package, bund foam generator package, water/foam hydrant/monitor, sprinkler control assembly and MJC assembly.

All hydrocarbon storage tanks are provided with foam systems and fixed spray nozzle rings, actuated by a fixed detection system and local manual release facilities.

The firewater system does not supply any process or other systems.

3.12 Waste Water Treatment

Note: For a detailed description of the Hammar IPF Waste Water Treatment System, refer to Operation and Maintenance Manual Volume 20 – Document 00251121DWPV50019.

The Waste Water System is designed to treat black water (sewage) and grey water from the living quarters giving due consideration to the environmental standards.

Black water defines sewage discharged from toilets.

Grey water describes the water and solids discharged from buildings that include washbasins and sinks, shower trays, washing machines, floor drains and kitchen equipment. The grey water is routed to the Grease Interceptor Package prior to entry into the 22 m^3/day Sewage Treatment Package (STP).

A separate facility is provided to treat or collect laboratories wash water that may contain toxic chemicals.

The Waste Water Treatment System includes:

- Grease interceptor package
- Sewage sump and pumps
- Sewage treatment package with sludge pumps
- Irrigation sump and pumps

4.0 PROCESS SYSTEMS OVERVIEW

4.1 Inlet Manifold

Note: For a detailed description of the Hammar Crude Inlet Manifold, refer to Operation and Maintenance Manual Volume 12 – Document 00251121DWPV50011.

Produced fluids are routed to the Hammar IPF from adjacent DGS and routed through four offsite interconnecting pipelines that connect at the well fluid inlet manifold. Locked closed (LC) double block and bleed (DBB) isolation valves are provided in the produced fluid inlet manifold to maintain segregation into four (4) separate identical production trains.

The inlet manifold includes eight (8) SDVs (one per inlet flowline and one per train) that are responsible for isolating each crude oil processing train and each flowline. This system allows the operators to align different flowlines with different trains, depending on the capabilities, maintenance status or operational requirements.

The inlet manifold is designed so that flowline number one can only be aligned with train two if train one is offline, and can only be aligned with train three if trains one and two are offline. This applies to all flowlines and all the trains for Hammar IPF.

Note: The system has not been designed such that one train is capable of being aligned with multiple flowlines.

4.2 Crude Oil Separation, Dehydration and Desalting

Note: For a detailed description of the Hammar Crude Oil Separation, Dehydration and Desalting System, refer to Operation and Maintenance Manual Volume 12 – Document 00251121DWPV50011.

The purpose of the Crude Oil Separation, Dehydration and Desalting System (Units 200/210) is to receive well fluids from Hammar Degassing Stations (DGS) through four (4) individual production trains, de-gas, de-water, de-salt and cool the crude oil to meet storage and export specification.

The crude processing/separation trains (Train 1/2/3/4) in the Hammar IPF each contain the following equipment:

- 1st stage separator (three-phase horizontal separator)
- Crude/crude oil exchanger (shell & tube exchanger)
- Crude oil heater (direct fired helical coil)
- 2nd stage separator (three-phase horizontal separator)
- Charge pumps (single stage centrifugal pump)
- Dehydrator (electrostatic, two-phase horizontal separator)
- Desalter (electrostatic, two-phase horizontal separator)

- Crude oil rundown cooler (air fan cooler)
- Wash water heat exchanger (plate & frame exchanger)
- Wash water recirculation pumps (single stage centrifugal pump)

Produced fluids from the manifolds at the existing adjacent DGS are directed to the 1st stage separators via buried pipelines.

The 1st stage separators are horizontal three-phase separator with proprietary internals to assist separation. It is designed to operate at two different pressures referred to as HP (36.5 barg) and LP (15.2 barg).

The separated oil phase (with up to 10% vol. water) flows through the crude/crude exchanger via a level control valve located upstream of the 2nd stage separator. This maximises the operating pressure in the exchanger and heater to limit the amount of gas released. Crude oil is pre-heated from 77°C to 80°C (summer case) or from 50°C to 60°C (winter case) by exchanging heat with crude product outlet stream from the desalter. The crude/crude exchanger is provided with a bypass line, so it is possible to send the crude oil leaving 1st stage separator directly to the crude oil heater, i.e., during maintenance or in turndown conditions. This heated crude is routed to the crude oil heater for further heating. The crude temperature is raised up to 95°C (LP operating mode) or up to 108°C (HP operating mode) to give an operating temperature in the 2nd stage separator and downstream equipment of approximately 90°C. Compressor condensate and off-spec crude recycle are also mixed with heated crude oil after the level control valve.

The control and shutdown of the crude oil heater are via the unit local control panel (LCP) with interfaces to the ICSS.

The heated crude from crude oil heater is routed to the inlet of the 2nd stage separator for further oil-gas-water separation. The 2nd stage separator operates at 2.8 barg. The crude oil from the 2nd stage separator (containing up to 5% vol. water), is pumped by the charge pumps (2 x 100%) and mixed with recirculated wash water through a mixing valve before flowing to the dehydrator. The vessel has split flow design, with a single inlet and two gas, oil and water outlets. The crude level in the vessel is controlled by means of a level control valve located downstream of the crude oil rundown cooler, so as to maximise the operating pressure in the dehydrator/desalter, crude/crude exchanger and rundown cooler to prevent any gas break-out before the degassing boot.

The charge pumps minimum flow protection is routed back to the 2nd stage separator.

The crude oil from the 2nd stage separator is further treated to achieve the crude specification, <0.5% vol. BS&W and a salt content of <30 ppm, in a two-stage electrostatic system, comprising a dehydrator and desalter, which are two-phase (oil-water) separators.

The dehydrator operates at a pressure of 6.7 barg. The wet crude enters into the vessel through an inlet distribution pipe and is evenly distributed below the oil water interface. The free water separates and is removed from the unit under level control and passes through the wash water heat exchanger where it preheats the fresh wash water going to the desalter, prior to being routed to the produced water tank.

The emulsion passes up through the grids, where electrostatic forces break the emulsion and purify the oil. The crude oil is collected in the top of the vessel, where it exits via a collection header. The crude is mixed with heated fresh wash water and mixed via a mixing valve, before entering the desalter, which operates in the same manner as the dehydrator. The crude oil is removed from the top of the desalter and wash water is removed from vessel under level control and fed back to dehydrator via the pre-heater.

The treated crude oil from the desalter is cooled in the crude/crude exchanger to 75°C/85°C (winter/summer), and further cooled by crude oil run down cooler to 50°C/60°C (winter/summer) to ensure that the flashed crude in the tank achieves the required crude TVP. The cooler has six (6) fans of which three (3) are variable pitch fans, and three (3) are fixed pitch fans. The temperature of the crude is controlled by modulating the variable pitch control on the air cooler fans.

An online BS&W analyser is provided downstream of the crude oil run down cooler to alert the operators if the crude goes out of specification.

The outlet of process trains are is normally routed to the crude oil degassing boot where any entrained gas in the crude oil is liberated to the ATM flare header. The degassing boot pressure floats on ATM flare header pressure. The treated oil from the degassing boot is routed into the crude oil storage tank by gravity.

In the event that the BS&W content of the crude to the storage tank exceeds 0.5% by volume, the operator is alerted and the crude is manually re-routed to the off-spec tank, until such time that the specification can be re-established. As with the crude oil system, a degassing boot is provided upstream of the storage tank and the liberated gas routed to the ATM flare.

Operating pressure on 1st stage separator is controlled by dedicated PIC, the PIC on the 1st stage separator acts in split range on two valves (PCVs), one on the line towards the HP flare system and one on the line towards the export HP compressor.

During HP mode operation, gas from the 1st stage separator is directed to the gas export line under pressure control, bypassing the HP compressors and free flows to the gas export line via the HP compressor discharge header. Gas for the HP fuel gas package is taken downstream of HP compressor discharge header. Gas for the LP fuel gas package and flare assist is taken off upstream of the PCV.

During LP mode operation, gas from the 1st stage separator is directed to the HP compressors under pressure control, compressed by the HP compressors and transferred to the gas export line. Gas for the HP fuel gas package is taken downstream of the HP compressor discharge header. Gas for the LP fuel gas package and flare assist is taken off upstream of the PCV.

During an upset in the HP export gas compression package or gas export system, the excess gas flows to flare system under pressure control.

The water phase from the 1st stage separator flows, under oil-water interface level control, to the produced water buffer tank via the degassing boot.

The gas from the 2nd stage separator is directed to the LP compressor, and the pressure in the separator is controlled by the compressor suction pressure with excess gas (during an upset in LP compression system) routed to the flare system. The water phase flows, under oil-water interface level control, to the produced water buffer tank via degassing boot.

Spent wash water from the dehydrator is routed to the produced water buffer tank via the desalter. Water from the desalter is returned to the dehydrator via the wash water recirculation pumps (2 x 100%).

Since the gas from the 1st stage separator and 2nd stage separator is wet, this may cause corrosion in the downstream piping facilities. To prevent this, chemical injection points are provided at the gas outlet streams of the two separators for the purpose of corrosion inhibition.

Each of the four (4) process vessels has an inlet nozzle for sand flushing water and outlet nozzles for sand slurry, and all the vessels are provided with internal sand flushing pipe work. However, initially, only the 1st stage separator sand flushing facilities shall be hooked-up. Sand removed in the vessel is routed to the desanding system.

The four (4) trains can operate with all in HP mode, all in LP mode or some in HP and some in LP mode.

An LP fuel gas take-off is provided at all trains, downstream of the 1st stage separator, but supplies to the LP fuel gas system shall be from at least two (2) trains operating in same pressure mode (LP or HP).

When one of the trains is out of operation for maintenance or due to equipment breakdown, the other oil treatment trains can still be operated as each is designed to operate on a standalone basis.

4.3 Wash Water System

Note: For a detailed description of the Hammar Wash Water System, refer to Operation and Maintenance Manual Volume 13 – Document 00251121DWPV50012.

The purpose of the Wash Water System (Unit 520) on the Hammar IPF is to provide pre-heated wash water for the removal of salt in the desalter (four trains).

Wash water for the desalting of the crude oil is supplied from the existing off-site river water supply.

The wash water system in the Hammar IPF includes the following main equipment:

- Wash water tank
- Wash water pumps
- Wash water heat exchanger
- Wash water recirculation pumps

The water stored in the wash water tank is de-oxygenated with oxygen scavenger from the chemical injection system.

Wash water is taken from incoming river water pipeline, injected with oxygen scavenger and routed to the wash water tank under level control. The oxygen content is kept below 10 ppb.

The wash water tank is blanketed with nitrogen to keep an oxygen free environment.

Wash water pumps (5 x 25 %) transfer the wash water for the desalting of the crude oil under flow ratio control to the wash water exchanger where it is preheated by transferring heat from the dehydrator produced water outlet. The pre-heated wash water is mixed with the crude coming from the dehydrator before it enters the desalter.

An oxygen analyser is provided on downstream of wash water pumps to monitor the oxygen level in wash water to the desalter.

4.4 Produced Water Treatment

Note: For a detailed description of the Hammar Produced Water Treatment System, refer to Operation and Maintenance Manual Volume 14 – Document 00251121DWPV50013.

The PWT system on the Hammar IPF treats produced water from the crude oil processing train, plus the wash water added for the desalting of the crude, and any additional capacity required for the returned water from the desanding flush of the separators, the backwash flow from the nutshell filters, pre-service flush water and skimmed oil from the IGF vessel.

The incoming produced water stream is routed to the produced water buffer tank via the produced water degassing boot, where any entrained gas in the produced water is liberated to ATM flare header. Reverse emulsion breaker is injected into the produced water upstream of degassing boot to facilitate oil droplet separation in produced water buffer tank. The tank is provided with a floating skimmer, to remove the oil layer that builds-up and is periodically transferred to the closed drain system by gravity under manual operation by a globe valve. The removal of the skimmed oil is visually monitored through a sight glass available upstream of the globe valve.

Produced water is routed to the PWT packages by the 5 x 25% produced water transfer pumps.

In the PWT package, oil is removed firstly in a CPI unit followed by a further clean-up in an IGF unit. These vessels are pressure vessels, which operate at 5.3 and 4.7 barg, respectively.

The gas for the separation process in the IGF is provided from the vapour space in the IGF vessel and make-up gas is provided from the MP fuel gas system. The IGF recirculating pumps (2 x 100%) circulate the produced water in the IGF for the operation of the gas educators.

The level control valve in the produced water outlet line controls the liquid level in the IGF and feeds the produced water to the nutshell filters.

The nutshell filters, 2 x 50%, provide the final polishing of the produced water to remove free oil in water to ≤20 ppm (volume basis) and achieve a suspended solids size to D95 ≤5 µm.

The treated produced water is routed to the treated produced water tank. Separated oil from the CPI unit is directly routed to the off-spec crude oil tank under level control on floating oil layer. Oil froth skimmed from the IGF unit (containing significant quantities of water) is fed to the produced water buffer tank under level control.

The CPI unit boots are fitted with sand fluidization facility to remove solid particle/sand, which is directed to the backwash settling tank.

The nutshell filters are periodically backwashed with produced water and scour gas and the backwash water carrying sludge, oil and scour gas is routed to backwash settling tanks. The backwash occurs periodically, and the facilities are designed for backwash of both nutshell filters twice a day based on design oil loading and purity of produced water.

One filter bed is backwashed first followed by the other. The duration of the NSF backwash is 16 minutes for one filter, a total 35 minutes for both.

Sludge from the nutshell filters backwash is removed from the backwash settling tank by truck for offsite processing.

The produced water buffer tank and IGF have gas blanketing systems to maintain injection water oxygen content level ≥10 ppb (by weight). The IGF also uses the fuel gas for floatation process.

The Hammar IPF is provided with the following:

- One PW buffer tank sized for four (4) process trains
- One treated PW tank sized for four (4) process trains
- Five 25 % PW transfers pumps

- Four PWT packages
- Four backwash settling tanks
- Two backwash pumps (2 x 100 %) for each backwashing settling tank

4.5 Crude Oil Storage and Export

Note: For a detailed description of the Hammar IPF Crude Oil Storage and Export System, refer to Operation and Maintenance Manual Volume 15 – Document 00251121DWPV50014.

The crude oil storage facility at the Hammar IPF consists of a crude oil storage tank (with degassing boot) and an off-spec tank (with degassing boot).

The crude oil and off-spec oil tanks are provided with blanketing systems for the in-breathing/out-breathing requirement of the tanks. The primary blanket gas supply is fuel gas, with nitrogen as back-up during start-up and shutdown conditions.

At the Hammar IPF, the crude and off-spec tanks are both common for all four trains with 6 x 25% crude export pumps and 5 x 25% off-spec pumps provided. The minimum flow recycle lines are both sized for four (4) pumps.

Crude from the crude oil storage tank is routed to the crude oil export pumps with a common minimum flow recycle line routed back to the storage tank via the degassing boot. An ultrasonic flow meter is provided on the crude export line and an online BS&W analyser is provided to monitor the crude flow rate and quality.

The stabilized crude oil is delivered at 24.5 barg into an existing crude oil export pipeline system.

The off-spec crude oil tank is used for the storage of off-spec crude from the oil treatment trains, and recovered fluids from the closed drain system. Off-spec crude is returned to the 2^{nd} stage separators by the off-spec pumps when there is sufficient capacity in the trains (up to a maximum of 20% of train capacity), or transferred to the crude oil storage tank if specification is achieved. The minimum flow recycle line is routed back to the off-spec crude oil tank via the degassing boot.

4.6 LP / HP Gas Compression

Note: For a detailed description of the Hammar IPF LP Gas Compression, refer to Operation and Maintenance Manual Volume 17 – Document 00251121DWPV50016.

Note: For a detailed description of the Hammar IPF HP Gas Compression, refer to Operation and Maintenance Manual Volume 16 – Document 00251121DWPV50015.

The Gas Compression Systems in the Hammar IPF comprise nine HP compressors and LP compressors with associated diesel engine drivers and auxiliary equipment.

The gas compression systems are installed to compress the process gas from the 1st and 2nd stage separators for export, with the separate LP and HP systems, operating in parallel.

During LP operating mode (1st stage separators operating at 15.2 barg (220 psig)), the HP compressors compress process gas from the 1st stage separators, while process gas from the 2nd stage separators is compressed by the LP compressors. The compressed gas from each compressor system is combined and routed to the gas export system with an export pressure of 41.3 barg.

During HP operating mode (1st stage separator operating at 36.5 barg (530 psig)), the 1st stage separator process gas bypasses the HP compressor system, and flows directly to the export line.

Gas from the 2nd stage separators is compressed by the LP compressors and routed to the gas export system. During this operating mode, the HP compressors are not operated. The LP compressors increase the process gas pressure from the 2nd stage separator to an export pressure of 35.4 barg.

For the Hammar IPF, the process trains may be operated in different modes. When the Hammar IPF is operating under these conditions the gas outlet from the trains operating in HP mode bypass the HP compressors and go directly to the gas export system, while those trains operating in LP mode are routed to the suction of the HP compressors for pressure boosting before going to the gas export system.

Any excess gas is flared in the event that the gas compression systems are being unavailable or restricted or if the off-site export system is unavailable.

Condensate from the compressor scrubbers is collected the condensate header and routed to off spec crude oil tank.

4.7 Gas Export and Metering

Note: For a detailed description of the Hammar Gas Export and Metering, refer to Operation and Maintenance Manual Volume 21 – Document 00251121DWPV50020.

The purpose of the Gas Export and Metering System is to measure the flow of gas from the Hammar IPF and to transport the gas to the main raw gas export line (not part of Weatherford scope).

The Gas Export and Metering System comprises:

- Gas Export Header
- Ultrasonic Flowmeter (FIT-009)
- Pressure Control (PIT/PCV-010)
- Export Pipeline Shutdown Valve (SDV-023)
- Pipework Isolation Valves, Non-return Valves and Instrumentation

Note: All personnel who are required to work on the Gas Export and Metering System should have read and have access to a copy of the MSDS (Material Safety Data Sheet) for chemicals present in the Gas Export and Metering System.

Note: Gas export and metering equipment tag numbers are prefixed by ‘3600’.

Produced gas from the Hammar IPF is routed to the gas export header. The gas export header is provided with take-off lines for HP compressor recycle gas and supply to the HP fuel gas treatment package.

A non-return valve is provided downstream of the two take-off lines to ensure that feed gas to the HP fuel gas treatment package can only be provided from the 1st stage separators or the discharge from the HP compressors.

Discharge from the LP compressors ties-in downstream of the non-return valve.

Gas flow through the gas export header is measured by an ultrasonic flowmeter with the flow measurement compensated with pressure and temperature input.

Pressure in the gas export header is controlled through pressure measurement and control with excess gas being routed via pressure control valve to the HP flare header.

Gas from the gas export header is routed to the gas export pipeline which is equipped with isolation valve, non-return valve, shutdown valve and a final isolation valve before entering the underground pipeline to the main raw gas export pipeline.

A “barred T” is provided upstream of the shutdown valve for future tie-in of pigging facilities.

5.0 INTEGRATED CONTROL AND SAFETY SYSTEM OVERVIEW

Note: For a detailed description of the Hammar Integrated Control and Safety System, refer to Operation and Maintenance Manual Volume 19 – Document 00251121DWPV50018.

The Integrated Control and Safety System (ICSS) provided at the Hammar IPF is a programmable logic controller (PLC) based system, which includes a Distributed Control System (DCS), Emergency Shutdown (ESD) system, Fire and Gas (F&G) System, and Closed Circuit Television (CCTV) system.

The ICSS is designed to provide safe and efficient operation of the Hammar IPF.

- Distributed Control System (DCS): The DCS is designed to provide monitoring and control of Hammar IPF Process and Utility Systems. The DCS is also the operator interface for the other elements of the ICSS
- Emergency Shutdown (ESD) System: The ESD element of the ICSS is a Safety System designed to reduce the risks to personnel by monitoring the Process and Utility Systems, and safeguard Hammar IPF from a dangerous event as a result of an upset. It also includes all executive actions (outputs), closure of emergency shutdown valves, opening blowdown valves and stopping pumps via their control panels
- Fire and Gas System (F&G) System: The F&G System is a safety related detection system, which is designed to monitor all the designated fire zones on the IPF for the presence of fire, heat, smoke or flammable gases using field detectors, generate alarms, and initiate actions as required to make the facility safe
- Closed Circuit Television (CCTV) System: A CCTV System has been installed around the plant and the perimeter fence for monitoring the flare operation and for security/safety. This allows visual monitoring from inside the Central Control Room (CCR) by operations personnel and in the Security Room/Guard House (GH) by security personnel. Two types of camera are provided for security: Pan, Tilt and Zoom (PTZ) cameras and fixed cameras. The flare stacks are monitored by PTZ cameras.

The ICSS is designed to provide safe and efficient operation of the Hammar facility. The ICSS complies with the requirements of all applicable Governmental and regulatory documents and the rules and regulations of Iraq, and uses the latest edition of IEC 61508 and 61511 as a general guideline.

6.0 OVERALL PROCESS START-UP

6.1 Black Start / Re-Start Philosophy

The general basis for the black start philosophy is to first start-up only the minimum number of utility and process systems required to produce enough HP fuel gas to start-up one Gas Turbine Generator. Power to these systems during the start-up process will be supplied by the Emergency Diesel Generator and Black Start Generator (for gas turbine auxiliaries). Once power is being generated by one Gas Turbine Generator, both diesel generators can be shut-down and other utilities (including other Gas Turbine Generators in case of Hammar IPF) and process systems can be started.

6.2 Utility Start-up

The following utility systems need to be started up first. These utilities need to be started in the sequence given below. Power to these systems will be provided by the Emergency Diesel Generator; therefore, this is the first system that needs to be started.

- Emergency Diesel Generator
- Emergency Switchgear
- Emergency Lighting
- 230V AC UPS and 110V DC UPS
- HVAC for control room
- Instrument Air Package
- Firewater System (Firewater Jockey Pumps and Diesel Firewater Pumps (with diesel engine) ready to start on demand. Electric Firewater Pumps are not required to be connected to EDG)
- Nitrogen Generation Package (required for tank blanketing, flare header purging and crude oil heater snuffing/purging. It shall be in high capacity mode)
- Atmospheric Flare System
  - Atmospheric Flare pilots ignited (using LPG back-up system)
  - Atmospheric Flare KO Drum Pumps ready to start on demand
  - Atmospheric Flare Blowers ready to start on demand (Hammar IPF only)
- LP / HP Flare System
  - HP Flare pilots ignited (using LPG back-up system) (Hammar IPF only)
  - HP Flare KO Drum Pumps ready to start on demand (Hammar IPF only)
  - LP Flare pilots ignited (using LPG back-up system)
  - LP Flare KO Drum Pumps ready to start on demand

In addition, the following utility systems will need to be powered from the Emergency Diesel Generator (EDG). These utilities (along with some of the utilities in the list above) may be required to be operational whenever the main power generation is not available, even before initiating the start sequence (e.g. after plant shutdown).

- Potable Water System
  - Potable Water Generation Package
  - Potable Water Pumps
- Diesel Fuel System
- Service Water System
- Closed Drain System
- Oil Contaminated Open Drain System
- White Water Open Drain System
- Sewer System

6.3 Process Start-up (Black Start)

In order to produce HP fuel gas to start the Gas Turbine Generators, well fluids will need to be admitted to the plant. The philosophy is to start-up a single crude oil train with the minimum number of equipment items in operation necessary to produce process gas in the 1st Stage Separator. This will in turn allow production of HP fuel gas to feed the Gas Turbine Generators. This philosophy implies that the crude oil produced during the black start procedure will not be on specification and will need to be routed to the Off-spec Crude Oil Tank for later re-processing. This approach also requires that all the process gas produced in the 2nd Stage Separator and most of the gas from the 1st Stage Separator will be routed to flare. In order to minimise the production of off-spec crude oil and minimise gas flaring during start-up, the throughput to the crude oil train shall correspond to the maximum turndown (25 % capacity). This will produce more than enough process gas to supply the fuel gas requirements during start-up.

Each crude oil train can operate on one of two modes, HP or LP operating mode. In HP operating mode, the arrival pressure of the well fluids is high enough to allow the HP Fuel Gas Package to be fed directly with process gas from the 1st Stage Separators. In LP operating mode, the operating pressure of the 1st Stage Separators is too low to be fed to the HP Fuel Gas Package and the gas from the 1st Stage Separator first needs to be compressed by the HP Gas Compressors before being routed to the HP Fuel Gas Package. Therefore, it is recommended to start-up a crude oil train that can be operated on HP mode (although black start with a crude train operating in LP mode is also possible).

After the essential utility systems described in Section 6.2 have been started, the process system black start sequence can be initiated. The general black start sequence of process units are described in the following sections 6.3.1 to 6.3.4.

6.3.1 Initial Set-up

- Well fluids will be routed to the online selected crude oil train
- Process gas from the 1st Stage Separator will initially be routed to the flare system
- Any gas produced from the 2nd Stage Separator routed to the flare system
- Produced water separated in the 1st and 2nd Stage Separators, together with any produced water from the Dehydrator, will be routed to the Produced Water Buffer Tank
- The crude oil through the train will follow the normal process route, i.e., 1st Stage Separators, Crude/Crude Exchanger, Crude Oil Heater, 2nd Stage Separator, Charge Pumps, Dehydrator, Desalter, Crude/Crude Exchange and Crude Oil Rundown Cooler. However, the Crude Oil Heater, Dehydrator, Desalter and Crude Oil Rundown Cooler will not be operating (the crude oil can be either put through the idle Crude/Crude Exchanger, Crude Oil Heater or bypass these equipment items)
- No wash water will be added
- The crude oil from the train will be off-spec and needs to be routed to the Crude Oil Off-spec Tank
- The Charge Pumps will be required to operate in order to get the crude oil from the 2nd Stage Separator to the Off-Spec Crude Oil Tank, and will be powered from the EDG
- Once well fluids are routed to the crude train, it may be required to inject corrosion inhibitor to protect the equipment and piping. Therefore, the Corrosion Inhibitor Pumps (for both the gas phase and liquid phase injections) of the Chemical Injection Package will need to be started

6.3.2 Fuel Gas Generation in HP Operating Mode

As mentioned above this is the preferred mode of operation for start-up.

- Once the flow of produced gas for the 1st Stage Separator has been established the HP Fuel Gas Treatment Package can now be started - the HP Fuel Gas Heater is switched on
- The surplus gas from the 1st Stage Separator will be routed to the Flare System for flaring

6.3.3 Fuel Gas Generation in LP Operating Mode

In LP Operating mode, the produced gas is not at sufficient pressure to produce HP fuel gas to run the Gas Turbine Generators. However, to operate an HP Gas Compressor (to increase the produced gas to the required pressure) it is necessary to produce MP fuel gas to supply the HP Gas Compressor.

- Once the flow of produced gas for the 1st Stage Separator has been established the LP Fuel Gas Cooler of the LP Fuel Gas Treatment Package is started (the

MP fuel gas to the HP Gas Compressors is not superheated; therefore, the LP Fuel Gas Heater is not required to be operating.)

- One of the HP Gas Compressors can now be started with the MP Fuel Gas available to supply pressurised produced gas for the production of HP Fuel Gas
- The HP Fuel Gas Treatment Package can now be started; with the HP Fuel Gas Heater switched on
- The surplus gas from the 1st Stage Separator will be routed to the Flare System for flaring

6.3.4 Electrical Power Generation and Balance of Plant Start-up

- The Black Start Diesel Generator, which provides electrical power to the Gas Turbine Generators, is started
- One of the Gas Turbine Generators can now be started. Once the GTG is running, and reached stable operating parameters to supply electrical power to the plant, the Black Start Diesel Generator can be stopped
- The Nitrogen Generator Package needs to be switched to low capacity mode (before stopping the EDG)
- The switch over for the supply of electrical power from the Emergency Diesel Generator (EDG) to the Gas Turbine Generator (GTG) can now be carried out
- The rest of the plant can now be started as per Start-up Operating Procedures

6.4 Process Re-start After Process Shutdown

In the event of a process shutdown, the HP Fuel Gas supply will stop; leading to the shutdown of the Gas Turbine Generators which will then not be available to re-start the plant. Therefore, the black start procedure described above needs to be followed to start-up the plant.

In case of process shutdown, the Emergency Diesel Generator will start automatically and the essential utility systems described in Section 6.2 will continue running powered by the EDG. However, if the process shutdown is caused by one of these utility systems (e.g., Low-Low instrument air pressure, High-High Flare KO Drum level, High-High HP Fuel Gas KO Drum Pressure, High-High HP Fuel Gas KO Drum level, etc.), the condition causing the process shutdown needs to be corrected and the affected utility systems restarted before the plant restart procedure can continue.

Similarly, before being able to start the process start-up procedure, all the process equipment described in Section 6.3 needs to be available and ready to be started. If the process shutdown is caused by one of these items of process equipment, the condition causing the process shutdown needs to be corrected before the procedure can continue. For instance, if the process trip is caused by High-High oil level in the 1st or 2nd Stage Separator (for Rafidiya or Zubair IPFs), the level in the vessel first needs to be reduced to its normal value by sending the oil either to the Off-Spec

Crude Oil Tank (via the Charge Pumps) or to the Closed Drain Vessel (after depressurisation through the PSV bypass). In the same way, if High-High pressure in either the 1st or 2nd Stage Separator causes a process shutdown, the affected vessel first needs to be depressurised to normal operating pressure through the PSV bypass.

For Hammar IPF any of these initiating events which only cause a TSD of the affected train and the rest of the plant can continue operating, unless only one train is in operation when the event occurs, in which case a PSD will occur.

There are a few causes of process shutdown that require special consideration, since the procedure to take the operating conditions to their normal values after a process shutdown is not immediately evident. These cases are described in the following sections 6.4.1 to 6.4.3.

6.4.1 Crude Oil Tank High-High Level

High-high level in the Crude Oil Tank will trip all crude oil trains leading to a process shutdown. After this, it is not possible to restore the level in the Crude Oil Tank to its normal value before starting the plant, since the Crude Oil Export Pumps are not connected to the Emergency Diesel Generator (EDG produces power a 400 V, while Crude Oil Export Pumps require power at 6.6 kV).

In order to restart the plant following a process shutdown caused by High-High level in the Crude Oil Tank, the general procedure described in Section 6.2**Error! Reference source not found.** and 6.3 needs to be followed, sending the off-spec crude oil produced to the Off-Spec Crude Oil Tank. Once one or more Gas Turbine Generators are started and synchronised, the Crude Oil Export Pumps can be started to export crude oil and reduce the level in the Crude Oil Tank. After the rest of the crude oil train is started and on-spec crude oil is produced, the train can be lined-up to the Crude Oil Tank.

6.4.2 Produced Water Buffer Tank High-High Level

High-High level in the Produced Water Buffer Tank will cause a process shutdown. Since there is no way to dispose the untreated produced water to reduce the level in the tank before restarting the plant, the procedure described in Section 6.2**Error! Reference source not found.** and 6.3 needs to be slightly modified.

In this case, the procedure previously described needs to be followed, with the exception that water is not removed from the 1st or 2nd Stage Separators. These vessels will work as two-phase separators, separating gas from liquid only. The liquids, which will be a mixture of crude oil with up to 50% water, will be sent to the Off-Spec Crude Oil Tank. The procedure described in Sections 6.2 and 6.3 continues normally until one or more Gas Turbine Generators are started and synchronized. At this point, one or more Produced Water Treatment Trains can be started to process the content of the Produced Water Buffer Tank and reduce its level to a normal value. Once this has been accomplished, the water outlet from the

1st and 2nd Stage Separators can be routed again to the Produced Water Buffer Tank and the train start-up procedure can continue as normal.

6.4.3 Off-Spec Crude Oil Tank High-High Level

High-High level in the Off-Spec Crude Oil Tank will trip all crude oil trains leading to a process shutdown. Before initiating the plant restart procedure described in Sections 6.2**Error! Reference source not found.** and 6.3, the level in the tank needs to be reduced to its normal value. The only way to achieve this is to send the tank content to the Crude Oil Tank (via Off-Spec Oil Transfer Pumps). However, this will contaminate the on-spec product in the Crude Oil Tank, taking it out of specification. The off-spec product in the Crude Oil Tank cannot be re-processed in the crude oil trains and will be need to be exported.

Since it is not possible to restart the plant after a process shutdown caused by High-High level in the Off-Spec Crude Oil Tank without exporting off-spec crude, this situation needs to be avoided. It is recommended to maintain the level in the tank as low as possible. The high level alarm in the tank will be set to alert the operator before reaching the maximum level in the tank (after which plant restart without exporting off-spec crude oil is not possible). The high level alarm will be set so that the time between the alarm and the maximum level in the tank is enough to allow the operator to take corrective actions before reaching this critical level (e.g., reduce plant throughput).

7.0 OVERALL PROCESS SHUTDOWN

7.1 Emergency Shutdown System

The emergency shutdown (ESD) system is the primary protection to prevent and minimise, as far as practicable, the occurrence and consequences of hazardous incidents resulting from failures within process systems. No control function is performed by ESD.

The ESD system monitors the safety shutdown field devices in the facility and produces permissive to run output signals for safety shutdown valves, rotating equipment, MCC, switchgear, etc.

The ESD system aims to prevent escalation of an upset condition and to ensure minimisation of damage to the environment, safety of personnel and protection of investment.

In general, the ESD system is provided for all process facilities that are involved in the processing or utilisation of hydrocarbon streams and associated utility systems and are used to bring relevant process and utilities equipment or part of the plant to a safe condition, when the process or safety (fire and/or gas) variable exceeds the limit of acceptable value. These include major fire, explosion, hazardous gas leak, potentially dangerous process upsets, e.g., high temperature, low pressure, etc., prolonged loss of major utilities, and all other events the operators perceive justifiable to interrupt the plant operations immediately.

The ESD system performs all process shutdown and blowdown functions as defined by the cause and effect (C&E) matrix diagrams whilst the fire and gas system initiates shutdown and blowdown as defined by C&E matrix diagrams and isolation of the electrical equipment.

All executive actions by the ESD utilise hard wired input/output (I/O).

7.2 ESD System Design

The ESD system is part of the Hammar IPF ICSS that continues to function normally even if any of the other ICSS components fail. The ICSS systems include:

- DCS
- ESD
- F&G

The ESD system includes independent dedicated process instrumentation which, on activation, will initiate the relevant executive actions on SDVs and blowdown valves. As a minimum, the ESD system provides:

- Automatic shutdown and isolation of equipment/systems to provide protection for environment, personnel and facilities
- Equipment/system depressurising (both automatic and manually initiated)

The ESD system achieves the above the following:

- Automatic sensing of an abnormal process or equipment condition
- Provision of audible and visual system status information for relevant personnel
- Shutdown of heat sources and operating machines
- Automatic operation of ESD isolating valves for process or utility fluids
- Block in sections of the plant
- Isolate hazardous liquid and gas inventories from leak and ignition sources
- Automatically initiate depressurizing of the plant, conditional on ESD level

The ESD system is designed to respond to instrumentation that detects upset operating parameters outside the normal operating envelope of that equipment (high and low process trips) and automatically drive the plant to a safe condition.

In addition, the operator is able to manually initiate a partial or total shutdown of the Hammar IPF using pushbuttons located in CCR.

The actions of the ESD system isolates the affected equipment or system to prevent escalation that may lead to a loss of containment, an uncontrolled release of hydrocarbons to the environment, hazardous conditions for plant personnel or damage to process equipment. This is achieved by the combined action of SDVs and BDVs.

Confirmed detection of a fire event from the F&G system initiates a shutdown, isolating the specific process train/package/compressors with auto-initiated blowdown of the affected unit/area.

Confirmed detection of a hydrocarbon gas initiates a shutdown, isolating the specific process train/package/compressors with manual blowdown of the affected unit/area.

With the exception of the auto-initiated blowdown scenarios described above, the decision to depressurise/blowdown the isolated system/equipment is the responsibility of the plant operations. However, if the operator fails to take prompt remedial action within a given time frame, then the ESD system alerts the operator to initiate a blowdown.

The emergency blowdown valves are activated through the ESD system by pushbuttons in the CCR and also automatically in the event of a confirmed fire event and a confirmed hydrocarbon gas release, as described above.

A confirmed fire event occurs by the activation of two (2) fire detectors (2ooN) in the same process train/area. Similarly, a confirmed gas event occurs by the activation of two (2) gas detectors (2ooN) in the same process train/area.

Alarms generated by the ICSS (from the DCS) are provided to warn operators of upset operating conditions, equipment failure, hydrocarbon gas release and the operators should have sufficient time to effect a recovery. If the upset operation conditions are not resolved and escalate, the ESD system detects the condition and initiates a shutdown.

The location of the SDVs for individual unit/area blowdown volumes and have been determined from safety studies, regulatory requirements and good design practice as follows:

- Plant battery limits (SDVs are fail safe)
- Unit battery limits for each process train/compressor package/area that correspond with a fire zone
- At the interface between HP/LP systems and other areas as required.

The operator can manually initiate a shutdown using pushbuttons located in the CCR.

Generally, following an ESD, normal operations can only be resumed after operations have determined the cause of the event and have taken remedial steps so that the plant can be restarted without tripping again. The exception to this is some of the unit shutdowns that can be automatically reset up to three (3) times, so long as there is no potential for cascade shutdown.

In general, ESD resets are performed with a 'soft' reset switch located in the CCR.

Control valves are generally not be used for emergency shutdown and isolation, although in a few specific non critical cases this has been allowed

7.3 ESD System Failure Mode

All ESD input and output instrumentation is designed to be fail safe and fails to a de-energised state. SDV failure states are shown on the P&IDs, i.e., fail open (FO), fail closed (FC) or fail no change of position (FL).

SDVs fail close on loss of instrument air supply. Where it is not desirable for an SDV to close on loss of instrument air, a separate dedicated air accumulator is provided. The solenoid valves actuating SDVs de-energise to close the SDV.

Therefore, the SDVs close on loss of electrical power. BDVs fail open. The BDVs are powered by the ESD system that is connected to an UPS supply. This ensures that BDVs do not open simultaneously on loss of electrical power.

7.4 ESD System Override Facilities

The application and removal of ESD overrides during maintenance is controlled at supervisory level via a master key switch on the ESD console in the CCR at the common service panel. This safety feature ensures that inadvertent application or removal of the ESD overrides does not endanger plant and personnel.

Every ESD initiator and final element can be overridden at the common service panel. Every item is alarmed and logged by ICSS.

7.5 SDV Reset

Following an ESD, normal operations must only be resumed after operations have determined the cause of the event and taken remedial steps so that the plant can be restarted without being tripped.

SDVs remain in their safe position following an ESD and are manually reset from the CCR or at the LCP for packages.

7.6 Shutdown Levels

Four levels of automatic shutdown apply to the Hammar IPF, which follows the principles and hierarchy outlined below (in ascending order of importance):

- Unit Shutdown (USD)
- Train Shutdown (TSD)
- Process Shutdown (PSD)
- Emergency Shutdown (ESD).

7.6.1 Unit Shutdown (USD)

A unit applies to cases where a hazard is limited to, and can be contained by, the isolation or shutdown of an individual piece of equipment or a selected group of equipment. USD has no effect on the rest of the facilities in terms of safety. No automatic depressurisation or blowdown is necessary except where required due to equipment requirements, e.g., compressor.

As it is possible for the plant to continue operating on loss of the compressors, they are considered as units and are, therefore, covered by an USD, as it is possible.

A USD includes the shutdown or isolation of all auxiliary equipment associated with the unit. If spare units are available, the control system may automatically start-up the standby units.

A USD initiates by in-field instrumentation and shuts down and isolates the affected equipment, a compressor or package, should tolerable excursions to the normal operating parameters be exceeded. SDV valves isolate the affected equipment/package/compressor, to prevent relief valves from lifting, such that in the event of loss of containment the hydrocarbon inventory released to the environment or potentially exposed to a fire is depressurised.

All executive actions are defined in individual process unit C&E matrix diagrams.

USD is initiated by the following:

- A PSD
- USD pushbuttons in CCR (specific packages only)
- Process trip for the specific equipment, e.g., oil transfer pump suction PALL

- Signal from higher levels of shutdown.

Confirmed detection of a fire event from the F&G system in compression area initiates a USD of all compressors, with auto-initiated blowdown. Confirmed detection of a hydrocarbon gas release gas in compression area initiates a shutdown and following a configurable time delay automatically initiates blowdown of the compressor package.

7.6.2 Train Shutdown (TSD)

A TSD is applicable to the four process trains and four PWT trains on the Hammar IPF.

A TSD initiates closure of all inflows to the train and isolates the train to be shut down from the other trains. The other three trains continue to operate.

A TSD signal is sent to the adjacent DGS, and operator intervention may be required to divert wells at the manifolds, or shut in specific wells.

The layout of the Hammar IPF is such that various process trains, compressors and the supporting utility systems are located in distinct plot areas. The distance between adjacent process trains allows each area to be considered a separate fire zone.

On initiation, a TSD actuates SDVs on all streams entering or leaving the train, stopping the fuel gas/heat input where applicable.

In addition, when a process train is shutdown, a number of LP compressors and HP compressors, if operating, need to be shutdown. The logic to identify the number of compressors to be shutdown is based on flow measurement from the crude flow rates for process train.

As a minimum, all streams entering or exiting a train are fitted with SDVs at the train battery limit to satisfy the requirements of a TSD shutdown.

A TSD is initiated by:

- A PSD
- Activation of the TSD pushbutton in CCR
- Major process trip within a specific train, e.g., 1st stage separator PAHH or LAHH, CPI PAHH etc.)
- Confirmed detection of fire event on specific train from the F&G system
- Confirmed detection of a hydrocarbon gas release from the F&G system (within a specific train).

A TSD of a process train initiates the following:

- Shut down of all SDVs isolating specific process train equipment

- Shut down of all electrical equipment (motors, transformers) associated with process train
- Stopping a number of compressors
- Manual blowdown permissive available via CCR pushbuttons for specific oil treatment train
- A TSD signal to the adjacent DGS.

A TSD of PWT train initiates the following:

- Shut down of all valves isolating a specific PWT train
- Shut down of all electrical equipment associated with relevant train
- Shut down of a produced water transfer pump
- Send a TSD signal to adjacent DGS.

TSD of a PWT train does not automatically require shutdown of a process train, as the produced water buffer tank capacity can be used to buffer the water flow and it may be possible to clear the deviation and restart the PWT train before the buffer tank reaches a high level.

Confirmed detection of a fire event from the F&G system initiates a TSD shutdown, isolating the associated train and auto-initiate blowdown, where applicable. Confirmed detection of a hydrocarbon gas release gas initiates a shutdown and following a configurable time delay automatically initiates a blowdown of the affected train.

In all other TSD scenarios, blowdown of the isolated process unit/area is operator initiated.

A TSD system reset is manually operated and only permitted once the shutdown cause has been locally checked and removed.

7.6.3 Process Shutdown (PSD)

A PSD initiates closure of all shutdown valves and blocks in the complete process facilities. No automatic depressurisation or blowdown is necessary, except where required due to equipment requirements. However, manual blow down permissives are available via CCR pushbuttons.

Due to loss of fuel gas, main power generation is shutdown and the Emergency Diesel Generator started.

Utilities, where appropriate, e.g., flare system, instrument air, nitrogen, potable water and HVAC, continue to function.

A PSD is initiated by:

- An ESD
- Activation of the PSD pushbutton in CCR

- A trip of the turbine generators (2oo4 voting)
- A high-high level in the flare KO drums
- A low-low pressure in the instrument air header (2oo3 voting)
- A high-high level in the crude oil storage tank
- A high-high level in the off-spec crude tank
- A PSD signal from the HP fuel gas package
- A confirmed MAC, fire or gas signal from the flare KO drum pit
- A confirmed MAC, fire or gas signal from the flare ignition package area
- A confirmed MAC, fire or gas signal from the crude oil storage tank
- A confirmed MAC, fire or gas signal from the firewater pump & storage area
- A confirmed MAC, fire or gas signal from the reception/export area.

A PSD initiates the following:

- Shutdown of all shut down valves, trains, equipment, packages and export
- Shutdown of the main power supply
- Shutdown of the fuel gas package
- Shutdown of the chemical injection packages
- Switches nitrogen generation package from low capacity to high capacity mode
- Manual blow down permissive available via CCR pushbuttons
- Auto start of emergency diesel generator (EDG)
- Close valves on incoming/outgoing pipelines
- Activation of TSD for train 1/2/3/4
- Send PSD signal to adjacent DGS
- Activation of relevant USDs including PWT, water injection, compression and minor utilities.

The instrument air system remains in operation.

The operator has a double action pushbutton to avoid a PSD being inadvertently initiated. Also, the PSD system reset is manually operated and only permitted once the shutdown cause has been locally checked and removed.

7.6.4 Emergency Shutdown (ESD)

An ESD initiates complete shutdown of the process facilities and utilities.

Activation of a higher level of shutdown automatically activates all lower levels of shutdown.

An ESD shuts down and isolates the Hammar IPF, achieved by cascading down through the lower shutdown levels.

An ESD stops the inflow to the plant from the upstream DGS, and isolates the individual process trains, vendor packages, and compressors bringing them to a safe condition ready for blowdown as well as alerting operators at existing facilities of the ESD.

Blowdown of the facility will be managed by the blowdown logic controller with blowdown initiated by operator intervention or auto-initiated.

An ESD is initiated by:

- Activation of the ESD pushbutton in CCR the (only manually activated by operator on positive confirmation of major incident in the Hammar IPF)
- A confirmed MAC, fire or gas signal from substation 1
- A confirmed MAC, fire or gas signal from the CCR
- A confirmed MAC, fire or gas signal from the power generation area.

An ESD initiates following:

- Automatic blow down of Train 1/2/3/4 1st stage and 2nd stage separators
- Activation of a PSD
- Shutdown of all utilities, except for nitrogen package, emergency diesel generator, HP/LP/ATM flare packages
- Closure of inlet flow line ESD valves
- Sends an ESD signal to adjacent DGS.

An ESD does not affect emergency services such as emergency lighting, instrument air systems (unless initiating event), nitrogen generation package, firewater pumps and firefighting systems. Activation of an ESD does not impede the operation of the following systems:

- Fire protection, suppression and deluge systems
- Potable water system
- Emergency power generation facilities and diesel supply.

The operator has as a double action pushbutton to avoid depressurisation being inadvertently initiated. Also, the ESD system reset is manually operated and only permitted once the shutdown cause has been locally checked and removed.

Once depressurisation is initiated, it cannot be stopped by the system or human intervention.

An initiated process depressurisation is completed, verified and the safety conditions confirmed prior to reset.

Spurious activated signal for process depressurisation is avoided by the use of appropriate control system configuration and proven technologies.

The emergency power generation and firefighting facilities are dependent of the instrument air supply.

7.6.5 Shutdown Interlock / Reset

To avoid the restarting of the process, in case of a shutdown, an interlock/reset philosophy is implemented:

- The alarms that caused the shutdown must be latched
- The first out alarm should be logged
- The devices involved in the shutdown must be interlocked
- The operator can check the alarms, even in the case that the conditions are normalised. Once the alarms have been cleared, the process can be started only if the operator resets the shutdown from the HMI. If the operator does not reset the shutdown, the process remains in the same status. Only with the reset can the devices involved in the shutdown lose the interlock status
- The reset applies at all levels of shutdown (unit reset, train reset, and process reset)

7.6.6 Process Trip Function

The cause and the effects of abnormal process operations are detailed in the cause and effect matrices. The need to install protective functions on equipment shall be considered by the process engineer on a case by case basis with a description of typical process trips described in the sections below.

In addition, abnormal process operations include a number of distinct scenarios during which the plant can continue to operate outside of the normal operating and control parameters for short periods of time. These controlled upset conditions include:

- Plant commissioning
- Start-up operations
- Manually initiated controlled shutdown operations
- Restart after shutdown for maintenance
- Process control functions designed to accommodate short term excursions from normal operating conditions, i.e., PCV that discharge to flare if system pressure exceeds a pre-determined setpoint.

For some of the above scenarios, the trips may have to be put into override, and several levels (classes) are identified as required to allow the starting of the process, such as:

- Class A trips are always active

- Class B trips are active after a time delay from the start of a piece of equipment or condition
- Class C trips are active after they clear the first time, e.g., low-low level trips
- During maintenance the associated alarms must be overridden

The class of trip is listed on the C&E documents.

8.0 APPENDICES

8.1 Abbreviations

Abbreviations	Description
AG	Above Ground
AFFF	Aqueous Film Forming Foam
ATM	Atmospheric
BA	Breathing Apparatus
BMS	Burner Management System
BOPD	Barrels of Oil per Day
BS	British Standards
BS&W	Basic Sediment and Water
CC	Corrosion Coupon
CCR	Central Control Room
CCTV	Closed Circuit Television
CIP	Cleaning-in-Place
CPI	Corrugated Plate Interceptor
CSP	Common Service Panel
DBB	Double Block and Bleed
DCS	Distributed Control System
DGS	Degassing Station
ECR	Emergency Control Room
EDG	Emergency Diesel Generator
EMS	Engine Management System
ER	Emergency Response
ESD	Emergency Shutdown
ESDV	Emergency Shutdown Valve
FEED	Front End Engineering and Design
F&G	Fire and Gas System
GH	Guard House
GOR	Gas/Oil Ratio
GTG	Gas Turbine Generator
H_2S	Hydrogen Sulphide
HMI	Human Machine Interface
HP	High Pressure
HSE	Health, Safety and Environment
IBC	Intermediate Bulk Container
ICSS	Integrated Control and Safety Systems
IEC	International Electrochemical Commission

Abbreviations	Description
IGF	Induced Gas Flotation
IPF	Initial Production Facility
IRCD	Injection Rate Control Device
ISO	International Standards Organisation
ISSOW	Integrated Safe Systems of Work
JHA	Job Hazard Analysis
KO	Knock Out
LAN	Local Area Network
LC	Locked Closed (Valve)
LCP	Local Control Panel
LER	Local Equipment Room
LO	Locked Open (Valve)
LP	Low Pressure
LV	Low Voltage
MMS	Machine Monitoring System
MMscfd	Millions of Standard Cubic Feet per Day (at 1.01325 bara and 15°C)
MSDS	Material Safety Data Sheet
NFPA	National Fire Protection Association
NRV	Non Return Valve
NSF	Nutshell Filters
OC	Oil Contaminated
OWS	Operator Workstation
PABX	Private Automatic Branch Exchange
PAGA	Public Address
PCS	Process Control System
PLC	Programmable Logic Controller
ppm	Parts per million
PRV	Pressure Relief Valve
PSD	Process Shutdown
PTW	Permit to Work
PTZ	Pan Tilt and Zoom
PW	Produced Water
PWT	Produced Water Treatment
RO	Restriction Orifice/Reverse Osmosis
RPE	Respiratory Protective Equipment
RVP	Reid Vapour Pressure
SCADA	Supervisory Control and Data Acquisition

Abbreviations	Description
SDV	Shutdown Valve
SQV	Deluge Valve
STP	Sewage Treatment Plant
SWL	Safe Working Load
TLV	Threshold Limit Value
TSD	Train Shutdown
TSV	Thermal Safety Valve
TVP	True Vapour Pressure
UCP	Unit Control Panel
UG	Underground
UPS	Uninterruptible Power Supply
USD	Unit Shutdown
US	Utility Station
VSV	Vacuum Safety Valve

8.2 Document Reference

Document Title	Document Number
Hammar IPF Cause & Effect Matrix	00251121DPEQ00101 Rev 04 (10/10/13)
Instrument List - Hammar	00251121DIEH60084 Rev 00 (23/10/13)
Process Control & Shutdown Philosophy	00250621DPSP00104 Rev 00 (05/09/13)
Crude Oil Train Processing Narrative	00250621DPST60007 Rev 03 (06/11/13)
Process Basis of Design	00250621DPSH00001 Rev 01 (25/08/13)
Black Start / Re-Start Philosophy	00250621DPSP00105 Rev 01 (29/08/13)
Operation & Maintenance Manuals	
Volume 1 Zubair Field and Hammar Overview	00251121DWPV50000
Volume 2 Instrument Air System	00251121DWPV50003
Volume 3 Nitrogen System	00251121DWPV50002
Volume 4 Chemical Injection System	00251121DWPV50001
Volume 5 Diesel System	00251121DWPV50004
Volume 6 Potable Water System	00251121DWPV50005
Volume 7 Service Water System	00251121DWPV50006
Volume 8 Flare System	00251121DWPV50007
Volume 9 Drain Systems	00251121DWPV50008
Volume 10 Fuel Gas System	00251121DWPV50009
Volume 11 Power Generation	00251121DWPV50010
Volume 12 Crude Oil Separation, Dehydration and Desalting	00251121DWPV50011
Volume 13 Wash Water System	00251121DWPV50012
Volume 14 Produced Water Treatment	00251121DWPV50013
Volume 15 Crude Oil Storage and Export	00251121DWPV50014
Volume 16 HP Gas Compression	00251121DWPV50015
Volume 17 LP Gas Compression	00251121DWPV50016
Volume 18 Firewater System	00251121DWPV50017
Volume 19 Integrated Control & Safety System	00251121DWPV50018
Volume 20 Waste Water Treatment	00251121DWPV50019
Volume 21 Gas Export and Metering	00251121DWPV50020